

AE582 Electrical Power Systems — Teaching notes (Case B focus)

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Purpose and learning outcomes

These notes prepare you to teach Case B, the RTG-powered outer-planet probe on slides 40 to 47, in about 40 minutes of a two-part lecture. A companion student study guide carries the same case with practice problems.

By the end of the Case B block a student should be able to:

1. Compute the average demand of a set of loads from standby power, active power and duty cycle.
2. Derate a source to end of life and explain why the budget is closed there.
3. Apply a design margin as a separate line and state the margin remaining.
4. Identify every operating state in which the load exceeds the source, and size the buffer energy for it.
5. Explain why a spacecraft can be power-positive on average and still need a battery.
6. Compare a continuous, limited source with an orbit-cycled solar source and name the operational lever in each.

Case B in five minutes

Case B has one idea: average power and peak power are separate questions, and an RTG answers the first but not the second.

- **The setup.** One Cassini-class RTG gives 285 W at launch and 210 W at end of life [2]. Five loads total 163.96 W on average and 312 W when all are active.
- **The average question.** $163.96 \text{ W} \times 1.20 = 196.75 \text{ W}$ is below 210 W. The budget closes with 13.25 W left.
- **The peak question.** 312 W is above 210 W. A buffer must supply 102 W for the 10-minute peak, which is 17 Wh.
- **The contrast with Case A.** In the PADRE case the battery carries the whole spacecraft through eclipse every orbit. Here the battery only covers peaks, because the source never switches off.

- **The lever.** Science and telecom duty cycles are schedule choices. Changing them closes or breaks the budget without touching hardware.

If time is short, teach slide 43 (the load profile against the RTG line), the worksheet on slide 44 and the first two tiles of slide 46. Those three carry the case.

Run of show

The full deck runs about 130 minutes at the pace below, with Case B taking 25 of them. The timings are my suggestion, not measured in class; adjust them to your session length.

Whole lecture

Block	Slides	Minutes	What must land
Roadmap and systems view	1–3	5	The EPS budget is an operations model in W and Wh
Requirements, power vs energy, modes	4–9	10	Budget by mode; peaks and averages are different numbers
Sources and solar cells	10–17	12	$1/r^2$, the I–V curve, design at end of life
Batteries	18–21	8	Depth of discharge trades mass against cycle life
Case A: PADRE model	22–25	12	Eclipse energy sizes storage
Array sizing, margins, concept check	26–29	8	Margins stay visible
Architecture, bus, DET and MPPT	30–35	12	Four jobs of an EPS; two ways to couple the array
Distribution, converters, battery control, derating	36–39	10	Losses are heat; faults must be isolated
RTG fundamentals	40–42	8	Continuous power, 5 % efficiency, large waste heat
Case B	43–46	25	Average closes, peak does not

Block	Slides	Minutes	What must land
System trade, make or buy, verification	47–51	8	Same requirement, different architecture
Team exercise and summary	52–54	12	Students choose and defend an architecture

Case B block, minute by minute

Minutes	Slide	Activity
0–3	43	Pose the question using the load profile. Ask: where does the load cross the RTG line?
3–5	44	Press Load nominal inputs . Explain the three columns: standby, active, duty.
5–13	44	Students compute in pairs: five averages, the total, EOL power, margin, buffer.
13–16	44	Press Check calculations on a volunteer's numbers, then Show solution .
16–21	45	Slider experiments: shorten the downlink, raise science duty, lower the retained fraction.
21–25	46	Discussion: what changed compared with Case A?

If you have only 15 minutes, skip the pair work and solve the worksheet at the board, then do one slider experiment.

Slide-by-slide notes, slides 40 to 47

Each slide below has one point to make and one question to ask the room.

Slide 40. When sunlight is the wrong resource: RTGs

- **Point:** an RTG trades the eclipse problem for a thermal and safety problem.
- **Say:** the output does not depend on Sun distance, pointing or eclipse. About 96 % of the thermal output remains after five years from decay alone [2].

- **Show:** on the decay chart, the electrical line falls faster than the thermal line because the thermocouples age as well. The electrical curve is indicative, not data.
- **Ask:** if the fuel loses only 4 % in five years, why does Case B lose 26 % of its electrical power?

Slide 41. An RTG is primarily a heat source

- **Point:** about 5 % becomes electricity and 95 % is heat [2].
- **Say:** 285 W electrical implies roughly 5.7 kW thermal. That heat must be radiated, or used to keep the bus warm.
- **Ask:** where on the spacecraft would you mount a 5 kW heater that also emits radiation?

Slide 42. Solar vs RTG with distance

- **Point:** array area grows with distance squared, while RTG output is flat.
- **Show:** 300 W needs about 1.3 m² at Earth, 34 m² at Jupiter and 114 m² at Saturn on the chart.
- **Caveat to state:** the chart scales by $1/r^2$ only and ignores low-intensity, low-temperature cell effects.

Slide 43. Case B workflow

- **Point:** the load crosses the RTG line only during telecom.
- **Show:** the green line at 210 W, the dashed average at 164 W, and the red area above the line.
- **Say:** the daily schedule drawn here is illustrative. It exists to show the shape, not to supply numbers.
- **Ask:** is this probe power-positive? Let them argue both ways before the worksheet.

Slide 44. Worksheet

- **Point:** students produce the four answers themselves.
- **Mechanics:** **Load nominal inputs** fills the givens and clears the answer cells. **Check calculations** marks each cell green within 2 % and red otherwise. **Show solution** fills everything.
- **Watch for:** pairs who average science as 65×0.25 and forget the 5 W standby.

Slide 45. Slider model

- **Point:** operations are a design variable.
- **Warn the class first:** this model is simpler than the worksheet, so its defaults read 170 W, 211 W, 3 % and 258 Wh. See "Three buffer numbers" below.
- **Do:** shorten the downlink from 2 h to 0.25 h and watch the buffer fall from 258 Wh to about 32 Wh.

Slide 46. What changes when generation is continuous but limited?

- **Point:** six tiles, of which three matter most: no eclipse recharge problem, operations become a lever, EOL still rules.
- **Ask:** why is adding a second RTG a different kind of decision from adding a strip of solar cells?

Slide 47. Same requirement, different architectures

- **Point:** the comparison table and the two sketches close the loop with Case A.
- **Say:** in LEO the battery cycles about 5500 times a year. On this probe it cycles about once a day.

Worksheet solution with instructor commentary

The worksheet has ten answer cells; the table gives each value, how it is obtained and what to say about it.

Cell	Value	Working	Comment for the class
Housekeeping average	80 W	Always on	Duty 1 means standby and active are the same number.
Thermal average	35 W	Always on	Heaters are a fixed tax on the budget.
Science average	20 W	$5 \times 0.75 + 65 \times 0.25$	The most common error is $65 \times 0.25 = 16.25$, which drops the standby term.
Telecom average	16.96 W	$8 \times 0.92 + 120 \times 0.08$	The largest load is the fourth-largest contributor.
Other average	12 W	Always on	

Cell	Value	Working	Comment for the class
Total average demand	163.96 W	Sum of the five	127 W of this is always-on load. Only 37 W is schedulable.
RTG EOL power	210 W	285×0.7368	The retained fraction is chosen so that the answer is the 210 W in [2].
Demand including margin	196.75 W	163.96×1.20	Margin multiplies demand. It does not reduce supply.
Remaining EOL margin	13.25 W	$210 - 196.75$	Positive, so the budget closes. True headroom over the best estimate is 28 %.
Required buffer energy	17 Wh	$(312 - 210) \times 0.1667$	A unit check: $W \times h = Wh$. Students often leave 10 minutes as 10.

The on-slide checker accepts values within 2 % of these, or within 0.02 in absolute terms for small numbers. A student who enters 197 W or 13.3 W is marked correct.

Two numbers worth putting on the board

- **127 W of 164 W is always on.** Housekeeping, thermal and other loads set a floor that no schedule can reduce. The margin is won or lost in the remaining 37 W.
- **The margin at launch would be 74 %.** Dividing 285 W by 163.96 W shows how misleading a beginning-of-life check is.

Three buffer numbers: 17 Wh, about 91 Wh and 258 Wh

The case produces three different buffer energies depending on how long the load is assumed to sit above the RTG line, and students will notice. All three are arithmetically correct; they answer different questions.

Buffer	Where it comes from	Assumption
17 Wh	Worksheet, slide 44	Only the 10 minutes with every load active counts: $(312 - 210) \times 0.1667$ h.
about 91 Wh	Not on a slide; derived below	The whole 1.92 h telecom pass counts, with a 10-minute science overlap inside it.

Buffer	Where it comes from	Assumption
258 Wh	Slider model defaults, slide 45	Science stays on for the whole 2 h downlink: $(340 - 211) \times 2$ h.

Where the 91 Wh comes from

The worksheet's own inputs imply a deficit for the entire telecom pass, which the worksheet does not ask about. With telecom active and science in standby the load is $80 + 35 + 5 + 120 + 12 = 252$ W, which is 42 W above the RTG.

$$E = 42 \times (1.92 - 0.167) + 102 \times 0.167 = 73.6 + 17.0 \approx 91 \text{ Wh per day}$$

This is a gap in the worksheet as written: 17 Wh would undersize the battery by a factor of five for the schedule its own duty cycles describe. I have left the worksheet unchanged and put this check in the student guide as an extension.

Why the slider model differs

The slider model lumps the always-on loads into a 140 W base, adds science (80 W) and telecom (120 W) as increments, and holds the peak for the full downlink. Its average is 170 W, not 164 W, because its telecom runs 2 h and its increments are slightly larger.

How to use this in class

Treat the three numbers as the teaching point. Ask which is right, then show that the answer depends on the schedule. The conclusion students should reach is that buffer sizing needs a list of every state above the source line, with its duration.

Options if you prefer one consistent number

- Change the worksheet's peak row to ask for the telecom-pass energy, with answer 91 Wh.
- Or lower the telecom active power to 78 W, so that telecom alone equals the RTG output and only the all-on peak needs a buffer.
- Or keep the worksheet and add one sentence on the slide stating that the 17 Wh covers the overlap only.

Tell me which you prefer and I will update the deck.

Common misconceptions

Six errors are likely in this case; each has a one-line correction.

Misconception	Why students hold it	Correction
"An RTG never runs out, so there is no end-of-life problem."	The half-life of 87.7 years sounds permanent.	Electrical output falls faster than the fuel decays because the thermocouples age. Case B loses 26 %.
"Average power closes, so no battery is needed."	Case A taught that batteries are for eclipse.	The load is 252 W or 312 W in two states. Both exceed 210 W.
"Average of a duty-cycled load is active power $\times$ duty."	It is the simplest formula.	Standby power runs the rest of the time. Science is 20 W, not 16.25 W.
"Peak load is the sum of every nameplate rating."	It is the safe-looking answer.	Peak is the sum of what is on at the same time. The schedule decides it.
"Margin means the source is derated by 20 %."	Margin and derating both reduce headroom.	Margin is added to demand. End-of-life derating is applied to supply. They are separate lines.
"Buffer energy is peak power $\times$ peak time."	They forget the source keeps running.	Only the part above the source counts: $(312 - 210)$ W, not 312 W.

Discussion questions with model answers

Use two or three of these on slide 46; each takes about two minutes.

- 1. Why is the battery on this probe smaller than you might expect for a 10-year mission?** It never has to carry the spacecraft through an eclipse. It covers at most about 91 Wh a day, while the PADRE battery in Case A supplies the whole eclipse load every orbit.

2. **The budget is tight at end of life. Would you add a second RTG or change the schedule?** Change the schedule first. A second RTG doubles the supply to 420 W but adds a large step in mass, cost, waste heat and safety effort. Cutting science duty costs data but no hardware.
3. **Which load would you cut first in a fault, and why?** Science, then telecom. Housekeeping and thermal are survival loads. With science and telecom in standby the load is 140 W, which leaves 70 W of surplus even at end of life.
4. **Where should the RTG be mounted?** Away from instruments that are sensitive to heat or radiation, often on a boom, with a clear view to space for its fins. Some of the heat can be routed to keep the bus warm.
5. **What would change if the probe went to Mars orbit instead?** Solar flux at Mars is 43 % of Earth's, so arrays are practical. The architecture returns to Case A, with eclipse-driven storage and no nuclear source.
6. **The margin is 28 % at end of life and 74 % at launch. What could the mission do with the early surplus?** Run higher science duty cycles early and reduce them as the RTG ages. The schedule becomes a function of mission time.

Full solutions to the student practice problems

The ten problems are in the student study guide; each solution below shows the working and the usual error.

#	Solution	Usual error
1	Science = $5 \times 0.60 + 65 \times 0.40 = 29$ W. Total = $163.96 - 20 + 29 = 172.96$ W. With margin = 207.55 W. Remaining = $210 - 207.55 = 2.45$ W. Closes.	Recomputing every row and slipping on telecom.
2	$P_{\text{avg}} \leq 210 \div 1.20 = 175$ W. Other loads total 143.96 W, so science ≤ 31.04 W. $5 + 60 D = 31.04$ gives $D = 0.434$, or 10.4 h per day.	Using 65 D instead of $5 + 60 D$.
3	$P_{\text{EOL}} = 285 \times 0.70 = 199.5$ W. Remaining = $199.5 - 196.75 = 2.75$ W. Buffer = $(312 - 199.5) \times 0.1667 = 18.75$ Wh.	Keeping 210 W in the buffer formula.

#	Solution	Usual error
4	$f = 196.75 \div 285 = 0.690$.	Dividing 163.96 by 285 and forgetting the margin (gives 0.575).
5	$102 \text{ W} \times 0.5 \text{ h} = 51 \text{ Wh}$.	Using 30 as hours.
6	$C = 91 \div (0.90 \times 0.40) = 253 \text{ Wh}$. Mass = $253 \div 130 = 1.9 \text{ kg}$.	Multiplying by depth of discharge.
7	Thermal = $285 \div 0.05 = 5700 \text{ W}$. Waste = 5415 W .	Taking 5 % of 285 W.
8	$P_{\text{EOL}} = 2 \times 210 = 420 \text{ W}$. Margin = $420 \div 163.96 - 1 = 156 \%$. Peak 312 W is below 420 W, so no buffer for this peak. Costs: mass, cost, double waste heat, safety effort.	Saying a battery is never needed. A small one is still wanted for transients and faults.
9	Average = level $\times$ fraction of time. Heaters: $35 \text{ W} \times 1$. Transmitter: $8 \times 0.92 + 120 \times 0.08 = 16.96 \text{ W}$.	Comparing 120 W with 35 W directly.
10	Telecom-active state is 252 W and all-active is 312 W. Both exceed 210 W, so the source cannot supply them whatever the average margin.	Arguing only from the 312 W state.

Problems 1 to 5 are routine. Problems 6 to 8 link to earlier parts of the lecture (battery sizing on slide 19, RTG efficiency on slide 41). Problems 9 and 10 test the two ideas the case exists to teach.

Assessment ideas and marking guide

Three short assessments fit this case: an exit ticket, a homework variant and an exam question.

Exit ticket (3 minutes, end of the block)

"State one reason a spacecraft that is power-positive on average can still need a battery." Accept any answer that names a state where instantaneous load exceeds source output.

Homework variant (about 45 minutes)

Give the worksheet with new inputs: science active 90 W at duty 0.20, telecom active 150 W at duty 0.10, retained fraction 0.72, margin 25 %. Ask for the four answers plus the telecom-pass buffer.

Quantity	Answer
Science average	$5 \times 0.80 + 90 \times 0.20 = 22 \text{ W}$
Telecom average	$8 \times 0.90 + 150 \times 0.10 = 22.2 \text{ W}$
Total average demand	$80 + 35 + 22 + 22.2 + 12 = 171.2 \text{ W}$
RTG EOL power	$285 \times 0.72 = 205.2 \text{ W}$
Demand including 25 % margin	214.0 W
Remaining EOL margin	$205.2 - 214.0 = -8.8 \text{ W}$. Does not close.
All-active peak	$80 + 35 + 90 + 150 + 12 = 367 \text{ W}$
Telecom-only state	$80 + 35 + 5 + 150 + 12 = 282 \text{ W}$, a 76.8 W deficit for 2.4 h = 184 Wh

The variant is built so that the budget fails. Ask students to propose one schedule change that closes it and to show the new numbers.

Exam question (10 marks)

Use the nominal case with one changed input, such as problem 3.

Marks	Awarded for
3	Correct load averages including standby terms
1	Total average demand
2	End-of-life power and demand with margin
1	Remaining margin with the correct sign and a statement of whether the budget closes
2	Buffer energy with correct units

Marks	Awarded for
1	One sentence explaining why average closure does not remove the need for storage

Give follow-through marks when an early arithmetic slip is carried correctly through later steps.